

Music-Gen — Cycles 10-12

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Music-Gen — Cycles 10-12

Abstract

Cycles 10-12 turned the campaign’s rules-and-generation spine from validated components into a live end-to-end deterministic generation pass and hardened the ledger surface that everything sits on. Cycle 10 was the fork 00b3ae64444c fanout with three clones: **M-GEN-1/first-generation** sampled one rule per rule_type from the 28-row M-RULES-1 ledger by SHA-256 tiebreak (no PRNG), assembled a 30 s three-Part MusicXML score, exported to MIDI through the M-SCORE-1 bridge, rendered bare via fluidsynth and effects-layered via the pinned Daw-Dreamer chain, measured the M-TEX-1/panel across the bare↔effects pair, and scored with M-HEUR-1 + the M-EAR-1/preparation CORN head under an explicit uncalibrated-labels sentinel; **M-INGEST-1/breadth-second-seeds** ran the eight-stage pipeline end-to-end on two additional on-disk seeds (seed_mid_50s, synth_060s) with 24/24 SHA-256 byte-determinism per seed, honest per-stage failure reporting, and a candid corpus-limitation /medium verdict for the absence of any non-synth audio; and a third clone tightened the ledger-writer schema so every future append validates against a single-source-of-truth module with status / narrative / nested confidence / explicit event_id required. Cycle 11 was a researcher pass framing the integration. Cycle 12 was the post-merge integration for fork 00b3ae64444c — the substantive adoption had already been done by the clones themselves through shadow-ledger emit into fanout concat, so the integration reduced to plan-row hygiene, two schema repairs on pre-hardening ledger rows, and a five-event capstone rollup emitted through the hardened writer, which caught real deficiencies in a first-draft event on its first live traffic and forced a corrected retry. At cycle-12 exit the workspace holds 191 ledger rows, zero promise_check ERRORS, and 413+ cross-branch integration test PASS lines across §1-§22 — including the load-bearing new sections §20 (schema hardening), §21 (M-GEN-1 SHA-256 anchors and PRNG-import grep guard), and §22 (breadth per-seed SHA-256 anchors × 2 seeds). Rated audio remains egress-blocked; the state machine that will unblock M-EAR-1 is IDLE and awaits its two-consecutive-media_ok=true trigger.

Introduction

By the end of cycle 9 the campaign had a rules-and-score axis that was validated in pieces: an M-RULES-1 ledger with 28 typed rules, an M-SCORE-1 bridge with round-trip byte identity, a fluidsynth + DawDremer render path, an M-TEX-1 panel with a refuse-to-aggregate contract, and an M-EAR-1 preparation chassis behind an uncalibrated-labels sentinel. What the campaign did not have was a live example that turned all of those pieces into a single deterministic generation. It also did not have a pipeline that had been demonstrated on more than one seed, and its ledger-writer helper had a recurring lesson — missing event_id, missing status, missing narrative, and non-canonical confidence — that had already been repaired twice in earlier cycles and would keep coming back until the validator was enforced at write time rather than at audit time. Cycles 10-12 addressed all three: first live generation, first pipeline breadth beyond one seed, and first hardened ledger writer that catches the recurring defects on the way in rather than on the way out.

Approach

Cycle 10 (fork 00b3ae64444c, three clones). Three parallel branches with disjoint file trees:

- **Clone 0 (M-GEN-1/first-generation).** `scripts/gen/sample_rules.py` groups the 28-row ledger by rule_type, computes SHA-256 over the canonical-JSON content of each candidate, and returns the lexicographically smallest hash per rule_type as the winner. Algorithm recorded on the manifest as `sha256_over_canonical_json_ascending` with `prng_used: false`; §21 of the cross-branch integration test greps every `scripts/gen/*.py` for `^(from|import) (random|numpy\.random|torch|secrets)` and fails on any hit. `scripts/gen/assemble_score.py` builds a three-Part music21 score satisfying the sampled rules' typed parameters. Render goes through the M-SCORE-1 bridge (`xml_to_midi`), fluidsynth with the pinned SF2 SHA 74594e8f...1cb0, and the cycle-9 DawDremer chain (Surge XT Chorus + Reverb + gain envelope) with determinism pins applied before any DawDremer import. Scoring writes `scoring_v1.json` with three blocks (M-HEUR-1 mess-scales; M-TEX-1/panel bare-vs-effects; CORN head prediction with `ear.calibration = "synthetic_labels_only"` sentinel). Provenance writes `provenance_v1.jsonl` — six canonical-order rows carrying `input_shas + output_shas + script + script_version` so the chain reconstructs from any intermediate step forward.
- **Clone 1 (M-INGEST-1/breadth-second-seeds).** `scripts/breadth/enumerate_seeds.py` swept corpus/seed/, corpus/ratings/, data/ingestion/seed/, and data/separation/synth_mix/ and confirmed on-disk that all 18 candidates are synth-derived (0 natural-recording seeds). Priority-order selection (non-synth first, then ≥ 30 s, then not the baseline) admitted the two seeds that maximise informational contrast within the available corpus: `seed_mid_50s` (50 s / 22 050 Hz mono / pure-sine content, exercising the sample-rate and upmix paths) and `synth_060s` (60 s / 44 100 Hz stereo / fluidsynth ground-truth, exercising the scaling axis on the baseline's content class). `scripts/breadth/run_seed.py` walks all eight stages, writes per-seed `stage_manifest.jsonl` with SHA / elapsed / diagnostic per stage, and drops all outputs into `data/breadth/<seed_id>/. Basic-pitch is invoked in its quarantined venv via subprocess with the cycle-6 environment pins passed through the child env.`
- **Clone 2 (ledger-writer schema hardening).** Extracted the ledger-event schema into a single-source-of-truth module `long_exposure/tools/_ledger_schema.py`, required status, narrative, nested confidence: {level, rationale, assessor}, and explicit event_id at write time, and added `tests/test_ledger_writer_validation.py` plus §20 of the cross-branch integration test. §20 pins the SSoT schema module identity so future edits cannot silently forget the contract.

Cycle 11 (researcher). Framed the integration cycle: no new research direction, no audit-level re-validation of the three clones' internal claims, one post-merge integration cycle whose job is to reconcile the workspace, run the validators, and close.

Cycle 12 (post-merge integration, worker). Two artefacts: `tools/stale/_repair_and_emit_fork_00b3ae64444c.py` (an `atomic-os.replace` ledger repair + five-event rollup emitter that also drives `promise_check` and both test suites and writes a JSON status report), and a rewritten `workspace-root merge_report.md` superseding the earlier `fork-f1bae241bde9` rollup. Two rows were added to `plan_of_record.md` for `M-INGEST-1/breadth-second-seeds/seed_mid_50s` and `M-INGEST-1/breadth-second-seeds/synth_060s` so `promise_check` accepts the pre-existing per-seed events on ledger lines 174 and 177. Two schema repairs on pre-hardening drift: line 160's `event_id` (a raw SHA-256 hex `3c9f2758...958d` produced by an ad-hoc emitter before clone 2's writer landed) was canonicalised via `uuid5(NAMESPACE_NIL, hex)` with the original hex preserved in `event_id_original`, and lines 179–184's `milestone_id`: `"M-TEST-1/writer"` (six round-trip test fixtures from clone 2's writer tests) were moved to the reserved namespace `_infra/ledger-writer-test-fixtures` with the originals preserved in `milestone_id_original`.

Findings

M-GEN-1/first-generation (cycle 10, clone 0) — validated/medium; parent M-GEN-1 rollup validated/medium

Five sampled rules:

rule_type	winner_rule_id	content_hash (first 16)	n_candidates
arrangement	rule_ 67d34b1c927ef33d	37dcaaf18bbf68ac	5
form	rule_ 84816f91e31e50c4	789c5c27825167fa	5
harmonic	rule_ 0271c7a9f3b5f606	(F major, [V, vii, iii, I, i, I, II, ii], cadence=none)	6
melodic	rule_ 09f340921fa2d258	—	6
rhythmic	rule_ 88b63bd5e771c045	—	6

Assembler summary: `active_parts=[drums]`, `key=F_major`, `meter=4/4`, `pattern_tokens=32`, `progression_len=8`, `seconds_per_measure=2.0`, `sections_dropped=29`, `sections_placed=4`, `tempo_bpm=120.0`, `total_measures=15`.

All six declared SHA-256 prefixes on disk match the worker's report exactly and were independently auditor-verified: `sampling_manifest.json` `faafc86ba79dccc2`, `generated.musicxml` `95d8671af26e7cf9`, `generated.mid` `f237dcfc75f5de94`, `bare_midi.wav` `5b6f608249ea72ac`, `effects_layered.wav` `d81089d39f31b5ca`, `scoring_v1.json` `011e7c90e1ab3c72`. A fresh re-run of `scripts.gen.assemble_score` under `PYTHONHASHSEED=0 OMP=MKL=OPENBLAS=1 /usr/bin/python3` into a scratch directory reproduced the `sampling-manifest` and `MusicXML` SHAs byte-identically.

Scoring: M-HEUR-1 mess-scales all finite in `[0, 1]` (melody 0.4358 / timbre 0.2938 / form 0.3029 / dynamics 0.9266); M-TEX-1/panel bare-vs-effects 8 keys with no aggregate (mel_l1_

db 16.76, spectral_centroid_rmse_hz 1798.62, rms_env_rmse 0.01057, lufs_m_rmse_lu 13.28, embedding_cosine_distance 0.0968); CORN head ear.prediction = 6 under the ear.calibration = "synthetic_labels_only" sentinel, surfaced in both §1 and §8.2 of the report so it cannot be mistaken for a musical judgment. provenance_v1.jsonl — six canonical-order rows, chain reconstructs from any intermediate step forward. The falsifiability hatch fired once and honestly: the first-pass scoring JSON leaked full paths and therefore drifted between runs; the fix stored basenames + SHA-256s.

M-INGEST-1/breadth-second-seeds (cycle 10, clone 1) — validated/medium

Both selected seeds passed 8/8 stages end-to-end (chunker → prepare-audio → M-CLASS-1 → M-SEP-1 htdemucs → M-TRANS-1 basic-pitch → M-SCORE-1 → render_bare_midi → M-TEX-1 panel). Byte-determinism across two independent runs: **24 / 24 SHA-256 anchors matched** (12 frozen contract artefacts × 2 seeds).

Panel numbers (original vs bare-MIDI) compared against the cycle-9 baseline:

seed_id	mel_l1_db	sc_rmse_hz	rms_env_rmse	lufs_m_rmse_lu	embed_cos	provenance
synth_030s (base-line, cycle 9)	9.906	2804.9	0.02759	2.682	0.1234	synth_ground_truth
synth_060s (this cycle)	10.755	2764.9	0.02887	2.843	0.1619	synth_ground_truth
seed_mid_50s (this cycle)	15.808	601.0	0.30918	20.837	0.1593	synth_seed_gen

Content discrimination is real (classifier: Sine wave $p = 0.9431$ vs Music $p = 0.8770$; panel RMS-env RMSE 11× different and LUFs-M RMSE 7.8× different between the sine seed and the ground-truth-family seed). The /medium cap is a corpus fact, not a stage-level defect: no non-synth audio exists on disk. Two “quiet passes worth calling out” are named honestly on the report — htdemucs’s energy skew on pure sines (correct-model-on-atypical-content, not a regression) and basic-pitch’s 5× over-count on the sine seed’s ~30 ground-truth notes (same octave-doubling artefact identified in cycle 8; the anti-pattern lock on re-attempting octave-suppression remains binding).

Family-disagreement recurrence. On synth_060s (same content family as baseline, longer duration), three of four numeric metrics track within 10 % while VGGish embedding cosine drifts 31 %. This is a second live datapoint validating the M-TEX-1/panel aggregation-refusal design commitment (the first was cycle 9 on M-TEX-1/stage-by-stage), and the mechanism is the same: VGGish’s mean_over_frames global summarisation is not scale-invariant when the underlying content distribution shifts even slightly.

Ledger-writer schema hardening (cycle 10, clone 2) — validated/high

Extracted the schema into `long_exposure/tools/_ledger_schema.py`; required status, narrative, nested confidence: {level, rationale, assessor}, and explicit event_id at write time; added `tests/test_ledger_writer_validation.py` and §20 of the cross-branch integration test pinning the SSoT module identity. The recurring cycle-7 lesson (append helper accepts events without event_id) is now enforced at write time rather than caught at audit time.

Post-merge integration (cycle 12, worker)

The substantive adoption had already been done by the clones themselves through shadow-ledger emit into fanout concat. Integration reduced to plan-row hygiene, two schema repairs, and a five-event rollup capstone:

#	milestone_id	status/conf
1	_infra/ledger-writer-test-fixtures	validated/high
2	_infra/repair-ledger-cycle10	validated/high
3	_plan/register-post-merge-integration-fork-00b3ae64444c	validated/high
4	_run/post-merge-integration-fork-00b3ae64444c	validated/high
5	_archive/integration-scratch-fork-00b3ae64444c	validated/high

Live proof-of-life of the hardened writer. The first-draft rollup events failed with `LedgerAppendError: ledger event schema validation failed on 3 field(s): missing required field 'status'; missing required field 'narrative'; confidence.rationale missing`. The tightening from cycle 10 caught real deficiencies in its own process on its first live traffic — exactly as designed — and after the fields were fixed the retry succeeded. The earlier draft also mis-called `append_ledger_event(event)` with the wrong arity (correct: `append_ledger_event(workspace, event)`), which failed loudly at the Python layer. Both are the kind of failure the hardened writer was built to make impossible to reach the ledger.

Verification at cycle-12 exit

- **Ledger:** 186 rows before integration → 191 rows after (5 rollup events; the shadow-ledger clone events had already merged in during fanout collapse).
- **promise_check:** rc = 0, **0 ERRORS**, 7 pre-existing WARNs unchanged (5 trailing-slash canonicalisation on old rows, M-EAR-1 parent with no events, and one orphan CORN-head feature-cache byproduct `data/ear/features/gen_first_gen_d81089d39f31b5ca.npz` from clone 0's ear scoring; trivial to adopt in a follow-up if desired).
- **tests/test_ledger_writer_validation.py:** rc = 0 (PASS).
- **tests/test_integration_cross_branch.py:** rc = 0, result: PASS (0 failures) across §1–§22 with 413 total PASS lines; §20 pins the SSoT schema module identity, §21 pins the M-GEN-1 byte-determinism SHAs and the PRNG-import grep guard (61 checks), §22 pins the breadth per-seed SHAs across two seeds.

Divergence, conflict, and overlap

None. All three cycle-10 clones self-verdicted validated; zero cross-branch content conflict; file trees disjoint (scripts/gen/*, scripts/breadth/*, and tests/test_ledger_writer_validation.py + long_exposure/tools/_ledger_schema.py). The M-TRANS-1/basic-pitch/octave-suppression anti-pattern lock (invalidated/high since cycle 8) held: clone 1 named it explicitly and did not re-attempt; clones 0 and 2 did not touch M-TRANS-1.

Discussion

Three things about this range are worth naming.

First, cycles 10-12 close the gap between “the pipeline is proven” and “the pipeline has run end-to-end on live inputs and produced a byte-deterministic artefact with a full provenance chain.” SHA-256 as the campaign’s universal tiebreak — content-derived rule_ids (cycle 6), content-derived event_ids (cycle 9), and now content-derived rule selection (cycle 10) — is what makes the generation legible under audit: every candidate’s hash is written to the manifest alongside the winner, and any future change to the ledger ordering, a new candidate rule, or an edit to any candidate’s content will change the sampler’s output in a fully-explainable way. A future switch to a PRNG-driven sampler is caught by the §21 grep guard. The pipeline is proven; the taste is not, and the report says so.

Second, the rule-composition incoherence surfaced by M-GEN-1/first-generation is not a bug in the pipeline — it is a real signal from a pipeline built on rules that were selected for content-hash determinism, not for musical coherence. The sampled arrangement rule silences pitched Parts; the sampled harmonic and melodic rules assume pitched Parts exist. Publishing this tension rather than patching it away is the falsifiability contract paying off again (cycle 8’s octave-suppression closure was the first live datapoint; this is the second). The right fix is a post-sampling M-GEN-1/rule-composition-constraint coherence gate that flags when the arrangement silences all pitched Parts or when form section granularity exceeds the target duration by more than 4×; the wrong fix is to put it inside the sampler, because that would break the SHA-256 tiebreak’s determinism.

Third, the cycle-10 ledger-writer hardening is the campaign’s answer to the recurring “append helper accepts an event without X” lesson that has already been repaired twice. Moving the schema into a single-source-of-truth module and enforcing it at write time turns a recurring audit-time defect into a caught-at-boundary error, and the fact that the cycle-12 integration’s own first-draft events were the first live traffic to trip the validator is the strongest possible evidence that the enforcement was necessary. _infra/repair-ledger-cycle10 cleaned up the pre-hardening drift (one raw-hex event_id, six M-TEST-1/writer test-fixture rows moved to the reserved namespace) with the originals preserved as event_id_original / milestone_id_original — an append-only ledger cannot lose the historical record, so the repair adds fields rather than editing them.

The uncalibrated CORN head remains the campaign’s biggest open credibility gap. The ear.calibration = "synthetic_labels_only" sentinel prevents the 6/7 rating on a drum-solo from being read as a musical judgment, and the M-INGEST-1/egress-ready-automation state machine will fire the retraining pipeline unattended the moment two consecutive fresh media_ok=true rows land. Everything downstream — CORN-head retraining, weight persistence at data/ear/corn_head_v1.pt with a feature-version guard, re-score of this cycle’s effects_layered.wav for calibrated comparison — is a straight-line consequence of that unblock.

Open Questions

- **M-GEN-1/rule-composition-constraint** — post-sampling coherence gate that flags arrangement-silences-pitched-Parts or form-granularity-too-fine-for-duration. Runs on the sampler's output, not inside the sampler; preserves the SHA-256 determinism.
- **CORN-head calibration.** When rated audio arrives, retrain the head on real labels, persist weights at `data/ear/corn_head_v1.pt` with a feature-version guard, and re-score the `cycle-10 effects_layered.wav` so 6/7 can be compared against a calibrated baseline.
- **Longer-duration M-GEN-1 targets.** `duration_s=60` or `90` so more of the sampled form rule's 128-measure structure lands. The current 30 s target was inherited from M-TEX-1/stage-by-stage; longer targets are cheap here.
- **Rules extraction per breadth seed.** Cheapest way to widen the M-RULES-1 corpus without new audio: run `scripts/rules/extract/from_score.py` over `data/breadth/{seed_mid_50s, synth_060s}/merged.musicxml`, emitting M-RULES-1/extraction/breadth-<seed_id> per seed.
- **SI-SDR-vs-mixture baseline on M-SEP-1.** Would catch pathologically-thin htdemucs splits on atypical content classes without re-training.
- **Split tests/test_integration_cross_branch.py by milestone** at ~890 lines. Cheap in scope; correctly deferred to a dedicated future cycle by cycle-10 workers so it does not entangle a substantive branch.
- **Adopt the CORN-head feature-cache byproduct.** `data/ear/features/gen_first_gen_d81089d39f31b5ca.npz` is a legitimate cycle-10 clone-0 side effect but is not in the declared artefact list; the next auditor pass should decide whether to adopt under M-GEN-1/first-generation or delete.
- **Documentation refinements** (auditor MINOR on M-GEN-1): update the report's total-check count from 343 → 413 and §21 count from 39 → 61; tighten the arrangement-silences-pitched-Parts claim to note that `ChordSymbol` MIDI realisations still fire even when the arrangement rule leaves pitched Parts otherwise empty. Not blocking.
- **CLAP-rung swap on the texture panel's embedding.** Orthogonal path to a more scale-invariant perceptual measure than VGGish's `mean_over_frames`; would revise the family-disagreement pattern now observed twice.
- **M-INGEST-1/egress-ready-automation firing.** Rated audio remains blocked; the state machine is IDLE and awaits its two-consecutive-media_ok=true trigger. Once it fires, `scripts/breadth/run_seed.py` is drop-in ready for the newly-arrived audio path (no code change required), and M-EAR-1 v0 training becomes eligible.

Appendix: Provenance

Cycle range: cycles 10-12. **Working directory:** `/home/user/long-exposure-runs/music-gen`. **Session references:** cycle 10 worker `32b9a84b-e739-4c2a-8bbd-3b80bee60cec`; cycle 11 researcher `cac1cf88-e8ab-45c7-9e09-cdde39869b1d`; cycle 12 worker `97cbc233-d4b1-426d-90a4-80947635eb6d`.

Sub-agent transcripts (fork 00b3ae6444c clones).

- Clone 0 (M-GEN-1/first-generation): researcher `c96fe124-a573-407b-9793-4dce6be05ae8`, worker `a9c6730b-5fc1-450f-8e2c-26bdb58cc27d`, auditor `179df4e2-bde2-4aa4-8bc5-6e58b7c16aaa`. Verdict VALIDATED, sub-milestone closes at `validated/medium`, parent M-GEN-1 rolls up at `validated/medium`.
- Clone 1 (M-INGEST-1/breadth-second-seeds): researcher `ed73c585-23ab-449c-9562-8a3ac46e5887`, worker `4d1aea55-0e21-43e0-ac6d-78fe877a7bb6`, auditor `44175d82-39e4-43ff-96a8-4084644a6b86`. Verdict VALIDATED at grade `/medium` under the brief's explicit downgrade rule for the on-disk corpus state.
- Clone 2 (ledger-writer schema hardening): verdict `validated/high`. Deliverables `long_`

exposure/tools/_ledger_schema.py, tests/test_ledger_writer_validation.py, §20 of the cross-branch integration test.

Deliverables on disk at cycle-12 exit:

- Code: scripts/gen/* (7 modules, interpreter-guarded, zero PRNG imports); scripts/breadth/* (interpreter-guarded, zero sidacar_nonfactor imports); long_exposure/tools/_ledger_schema.py (SSoT schema module).
- Data: data/gen/{sampling_manifest.json, generated.musicxml, generated.mid, renders/{bare_midi.wav, effects_layered.wav}, scoring_v1.json, provenance_v1.jsonl, render_manifest.json}; data/breadth/{seed_mid_50s, synth_060s}/* (full per-seed artefact sets) + {summary.tsv, seed_enumeration.tsv, determinism_baselines.txt}.
- Figures: docs/figures/{gen_first_generation_provenance.png, pipeline_breadth_panel.png}.
- Reports: docs/gen_first_generation_report.md (397 lines); docs/pipeline_breadth_report.md (281 lines).
- Tests: tests/test_ledger_writer_validation.py; tests/test_integration_cross_branch.py — 413 PASS / 0 FAIL, with §20 (schema hardening), §21 (M-GEN-1, 61 checks including PRNG-import grep guard + SHA-256 anchors + provenance-chain shape), §22 (breadth per-seed SHA-256 anchors × 2 seeds).
- Repair + rollout tooling: tools/stale/_repair_and_emit_fork_00b3ae64444c.py; workspace-root merge_report.md rewritten with the actual integration outcome.
- Plan of record: 2 sub-milestone rows added at cycle 12 for M-INGEST-1/breadth-second-seeds/{seed_mid_50s, synth_060s}.

Ledger state at cycle-12 exit: 191 events. Repair events: _infra/ledger-writer-test-fixtures, _infra/repair-ledger-cycle10. Line 160's event_id canonicalised via uuid5(NAMESPACE_NIL, "3c9f2758...958d") with event_id_original preserved. Lines 179-184 (six M-TEST-1/writer test fixtures) moved to the reserved namespace _infra/ledger-writer-test-fixtures with milestone_id_original preserved.

Environment stack unchanged since cycle 9: mscore3 3.2.3 headless; Python 3.11.15; numpy 1.26.4; music21 9.1.0; mir_eval 0.8.2; fluidsynth (Debian) with pinned SF2 74594e8f...1cb0; DawDreamer + Surge XT Effects.vst3 at /usr/lib/vst3/; basic-pitch 0.4.0 in workspace/basic_pitch_venv/; VGGish rung on the texture panel; M-EAR-1/preparation CORN head under the synthetic_labels_only sentinel. Single-thread BLAS pins throughout; DawDreamer determinism pins applied before import.

Rated audio. Still egress-blocked per corpus/CORPUS_STATUS.md. M-INGEST-1/egress-ready-automation will fire the retraining pipeline unattended when two consecutive fresh media_ok=true rows land in data/ingestion/egress_status.jsonl. Not this cycle's problem; the state machine has been on disk since cycle 8, IDLE, and its runtime state files remain correctly absent until the first live trigger.

Handoff to next cycle. The natural next research step is M-GEN-1/rule-composition-constraint as a post-sampling coherence gate — the smallest legible unit that raises M-GEN-1 above /medium without touching the SHA-256 tiebreak's determinism. The cheap follow-up is rules extraction over the two new breadth-seed merged MusicXMLs to widen the M-RULES-1 corpus. The pre-wired large unblock remains M-EAR-1 v0 training on rated audio; when it arrives the entire chain from egress-ready-automation through CORN-head retraining, weight persistence, and re-scoring is a straight-line consequence of the trigger.